Gender Bias in Automated Hiring — the Case for De-Identification

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Abstract

Hiring tools increasingly sort applicants automatically. A retriever pulls plausible resumes from a large pool. A re-ranker then puts them in final order. We audited 14 widely used re-rankers with counterfactual pairs: texts identical except for the candidate's name and pronouns. All 14 favored men more often for male-typed jobs than for female-typed ones. Which way a model leans by default is a property of that product, and cannot be known without testing it. In a simulated pool of 20 equally qualified candidates, one common pipeline gave men 79% of shortlist places, about four in every five. Removing names and pronouns before scoring removes the difference, because it removes the only thing that differs. Half-measures do not: deleting names alone leaves the direction almost perfectly consistent. We argue that de-identification is a cheap, auditable, less discriminatory alternative that deployers should be expected to adopt.

1 Why this matters

Software that sorts job applicants usually works in two steps [10]. A *retriever* scans a large pool and pulls out a few dozen plausible candidates. A *re-ranker* then reads each one against the job description and fixes the final order. The retriever is the clerk who pulls a stack of files; the re-ranker is the decision-maker who reads them and arranges them best-first. The people at the top of that order get looked at. The rest, in practice, do not.

This is not hypothetical. LinkedIn Recruiter returns a ranked list of candidates [9]. Indeed builds ranking systems to match job seekers to roles [8]. Google sells a job-search ranking service to employers [5]. Published resume-matching systems score a job description and a candidate document together, using exactly the kind of re-ranker we audit [7, 20].

Researchers have shown that the retriever carries gender bias [3, 11, 19]. The re-ranker has drawn far less attention, and no prior audit scores identical candidates across commercial ranking products. That is the wrong way round. The re-ranker has the last word, and it decides who appears at the top of the recruiter's screen.

Gender bias in neural rankers is not new. Rekabsaz and Schedl showed that BERT rankers surface more male-oriented passages than keyword search for neutral queries [14], and later added an adversarial fix [15]. Both papers ask what the retrieved text is about, not how two candidates who differ only by sex are scored. Audits of language models do ask that, and they disagree on direction. One study of large language models scoring resumes finds an advantage for women [1]; an embedding-based resume screener favors male-associated names [19]. We run the counterfactual test on the re-rankers that commercial hiring stacks actually use, and follow it through to who makes the shortlist.

Two legal regimes already cover this ground. In the United States, a neutral-looking practice that screens out one sex at a disproportionate rate is unlawful unless the employer shows business necessity. Even then it fails if a less discriminatory alternative exists and the employer declines it. In the European Union, systems that filter job applications are classed as high-risk. From December 2027, the organisations that deploy them owe duties to control that risk. Both regimes ask the same practical question: is there a cheaper, fairer way to do the same job?

2 What we did

Our test is a counterfactual pair: two versions of the same professional document. They are identical word for word, except for the candidate's name and pronouns. Any difference in score must therefore come from gender, because nothing else differs. We built these documents from 4 templates across 52 occupations. We varied 10 common American given names of each sex, taken from national birth-name records [16]. We used 4 ways of phrasing the search. That yields 8,320 counterfactual pairs per model.

We label each occupation by the share of women employed in it, from the federal labor-force survey [2]. 18 occupations are male-typed, 21 are female-typed, and 13 are close to evenly split. The evenly split group carries most of the legal weight, because nobody can claim those jobs require a particular sex.

We scored every counterfactual pair with 14 re-rankers, both commercial and open-source. Some models return coarse scores that give the two documents exactly the same number. We count those ties as ties, and never credit them to either side. The appendix lists every model, its vendor, and its tie rate.

To see what this does in practice, we simulated a pool of candidates for one job. It holds 20 documents, half with men's names and half with women's, identical in every other respect. Because the candidates are by construction equally qualified, a fair system should treat them interchangeably. We ran the pool through a common open-source pipeline and recorded who made the top three. Finally, we re-scored everything after stripping gender markers, at several strengths, to see which strengths actually work.

3 What we found

Identical candidates are not scored equally

All 14 models favored the man more often for male-typed jobs than for female-typed ones. The size of that gap varies; its direction never does. For male-typed jobs the middle model favored the man in 74% of comparisons, about three in every four. For female-typed jobs the middle model favored the man in only 33%. Nothing changed between the two documents but the name and the pronouns. The tilt is not random. It follows the same stereotype that would be unlawful in a human screener.

Where a model sits by default is a different matter. On jobs the labor market splits evenly, the middle model favored the man in 56% of comparisons, just over half the time. But the models themselves disagree: 8 of the 14 lean toward men on those jobs, and 6 lean the other way. A buyer

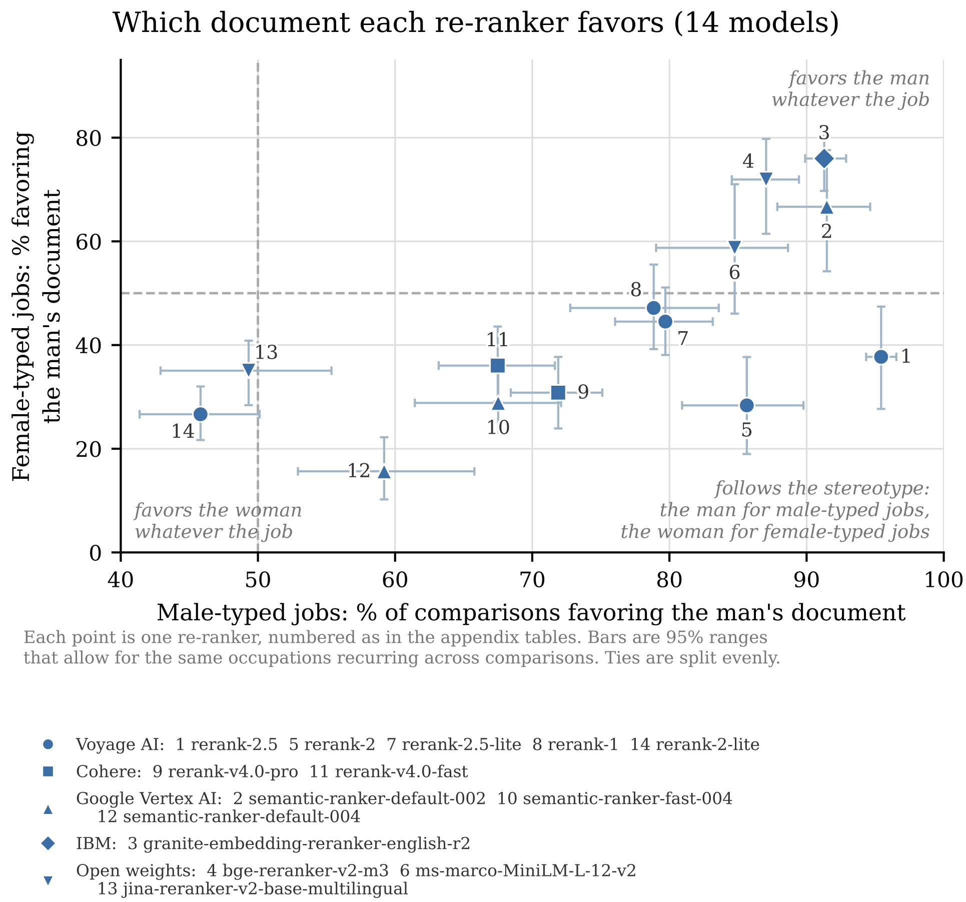

Figure 1: Each point is one re-ranker: how often it favored the man's document for male-typed jobs (horizontal) against how often it favored the man's document for female-typed jobs (vertical). A model that treated the pair alike would sit at 50% on both axes. Points are numbered as in the appendix tables, and the marker shape gives the vendor. Where a model scored the two documents exactly equal we split the tie evenly rather than crediting either side. Bars show the 95% range across occupations.

therefore cannot guess which sex a given product disadvantages. The direction is invisible from the outside, and knowable only by testing.

The re-ranker has the last word

A pool of 20 interchangeable candidates should produce a shortlist that is half men and half women. It does not. We expected the two stages to compound. A candidate must clear both, so two filters leaning the same way should skew the shortlist further than either alone. That is what happens where they do lean together. On male-typed jobs the retriever alone gave men 76% of the top three places, the re-ranker alone 90%, and the two in sequence 95% – worse than either stage on its own.

Where the two stages disagree, the effect reverses. On female-typed jobs the retriever leans the other way, giving men 45% of shortlist places. That pulls the pipeline back from the re-ranker's 70% to 67%. Across all jobs the two effects cancel, and the pipeline comes to resemble the re-ranker

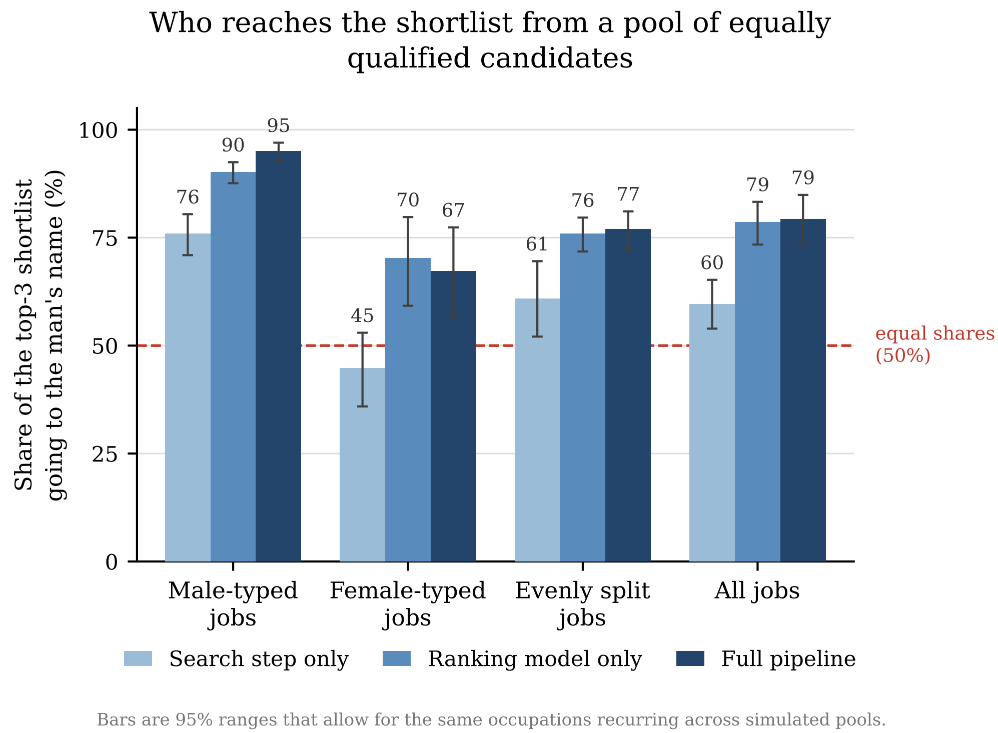

Figure 2: Share of top-three shortlist places going to men's names, in a pool of equally qualified candidates: under the retriever alone, the re-ranker alone, and the two combined.

alone: 79% against 79%. For the evenly split jobs it is 77%.

Favoring either sex among interchangeable candidates is a disparity, so none of these figures is the right answer. Two lessons follow. The pipeline cannot be read off either stage alone; what matters is whether the two happen to agree. And fixing the retriever achieves little on its own, because the re-ranker has the last word.

Removing the markers removes the gap, if you remove all of them

If both documents become the same text, any model must score them the same. Our full transform replaces the name and neutralises the pronouns, and every pair then ties. That is a check on the argument rather than a discovery, because identical texts cannot be scored differently. Half-measures are the interesting case: deleting names but leaving pronouns makes the average score difference *smaller*, from 0.036 to 0.033 on this model's own scale. But it makes the direction almost perfectly consistent: the man's document wins 99.7% of male-typed comparisons, up from 87%. Names carry a large effect that varies from pair to pair, while pronouns carry a small one that always points the same way.

So a half-transform is not half a fix. The obvious repair, rewriting pronouns as *they*, *them*, and *their*, works only if the rewriter understands grammar. English *her* is both possessive and object, so mapping it to one word leaves the pair differing by a word. That happens on 25% of pairs, and the model scores every one of them differently. A rewriter that checks each word's part of speech makes the two documents identical and keeps the text readable. Neither route costs accuracy: the right candidate came out on top 99.5% of the time, exactly as before.

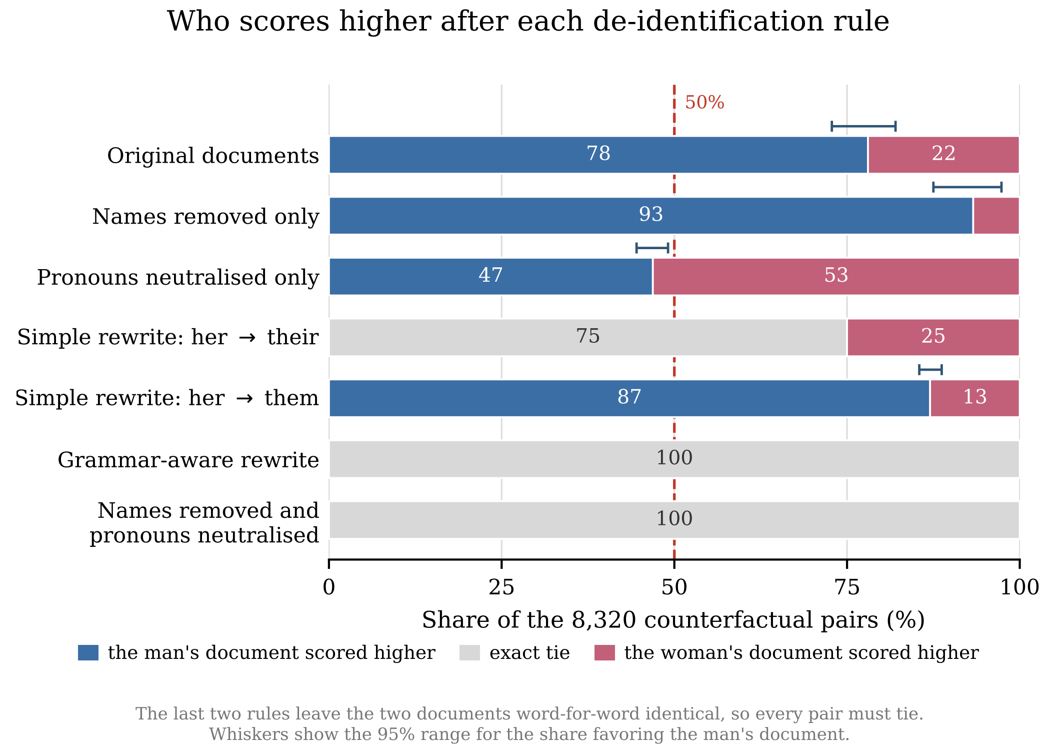

Figure 3: Who scores higher under each de-identification rule, across all counterfactual pairs. Each bar splits into the share favoring the man's document, the share scored exactly equal, and the share favoring the woman's document. The last two rules make the two documents the same text, so every pair ties; the simple *her* → *their* rewrite leaves a quarter of pairs still different. Job-category breakdowns are in the appendix.

4 What the law makes of it

Judged by the result, not the mechanism. United States law does not ask how a hiring tool reaches its decision. It asks whether a neutral-looking practice selects one sex at a disproportionate rate [4, 6].[1] So the number that matters is not the small score difference, but the shortlist. In our pool of interchangeable candidates, women took 21% of shortlist places. Men were shortlisted at a rate of 23.8% and women at 6.2%, a selection ratio of 0.26. That is about a quarter of the men's rate, far under the four-fifths level federal enforcement agencies treat as a warning line [17].

Our pool is a stylised illustration of that metric, not an applicant-flow analysis of a real employer's hiring. It shows what ranking does to a group of people the system itself cannot tell apart. Small, consistent nudges in score become a hard line between making the cut and missing it.

An employer cannot know the direction without testing. The direction of a model's default lean is a property of the individual product. A deployer therefore cannot know which sex a tool

[1]The rule is codified at 42 U.S.C. § 2000e-2(k)(1)(A). Clause (i) makes the practice unlawful unless the employer shows it is job related for the position in question and consistent with business necessity. Clause (ii) makes it unlawful anyway where the complaining party identifies an alternative employment practice and the employer refuses to adopt it.

disadvantages until it is measured. Two vendors' models, both marketed as accurate, can push opposite ways on the same job. Buying on trust is not a defence, and a counterfactual test of the kind we run costs an afternoon.

The evenly split jobs are the sharpest case. For a job the labor market divides evenly between men and women, no business-necessity story is available. An employer cannot argue that the work requires men. Yet the shortlist for those jobs was 77% male. The skew is not a feature of the work. It is a property of the software.

De-identification is a textbook less discriminatory alternative. Even a justified practice is unlawful if a less discriminatory alternative exists and the employer refuses it. De-identification qualifies. It costs almost nothing and can be read and checked by hand. Within the channels it covers, it drives the disparity to zero rather than merely shrinking it. The alternative must also be equally effective at the underlying job.[2] That is why we measured accuracy, and found it unchanged.

Europe arrives at the same place. The EU AI Act classes systems used to filter job applications as high-risk. It warns that such systems "may perpetuate historical patterns of discrimination, for example against women" [12]. The Act's design duties fall on the vendor, but an employer that only deploys the system still owes duties of its own. It must run the system as instructed, oversee it, and keep the input data relevant to the system's purpose.[3] A further duty to assess the effect on fundamental rights before use reaches public bodies and public-service providers, not ordinary private employers.[4] De-identification is a measure the deployer controls, applied at exactly that input boundary.

5 What to do about it

Anyone deploying two-step ranking over documents that describe people should remove gender markers from the documents and from the search queries, before either step runs. The specification below is short enough for a court, a regulator, or a compliance team to require and to verify.

1. **Replace names.** Substitute every personal name with a fixed placeholder such as "the candidate", or delete it. A real system needs a name detector rather than a word list, and that detector's coverage is itself an audit item.

[2] 42 U.S.C. § 2000e-2(k)(1)(A)(ii), read with § 2000e-2(k)(1)(C), which fixes the standard as the law stood on 4 June 1989. Under that law, cost and other burdens bear on whether a proposed alternative would be "equally as effective" in serving the employer's legitimate business goals [18].

[3] Annex III, point 4(a) and Recital 57 classify recruitment systems as high-risk. The requirements addressed to the system itself — risk management (Art. 9), data governance (Art. 10) and human oversight (Art. 14) — bind the provider, through Art. 16(a). The deployer's own duties are in Art. 26: to use the system in accordance with the instructions for use (Art. 26(1)); to assign human oversight to competent staff (Art. 26(2)); to ensure input data is relevant and sufficiently representative so far as the deployer controls it (Art. 26(4)); and, for an employer, to inform workers and their representatives before use (Art. 26(7)). Under Art. 113 as amended by Regulation (EU) 2026/1744, these obligations apply from 2 December 2027 [13].

[4] Art. 27(1), which is limited to deployers that are bodies governed by public law or private entities providing public services, together with deployers of the credit and insurance systems in Annex III points 5(b) and 5(c). Regulation (EU) 2026/1744 amended only Art. 27(4) and (5); the scope in Art. 27(1) is unchanged.

2. **Neutralise pronouns, by one of two routes.** The simplest is to replace every gendered pronoun with a single token. That is trivial to implement, but it produces ungrammatical text. The alternative is to rewrite them as *they*, *them*, and *their* with a tool that checks each word's part of speech. Both make the two documents identical, and the second reads normally. Do not use a plain find-and-replace for *they/them/their*: English *her* is both possessive and object, and ignoring that leaves a gender signal behind.

3. **Apply it at both steps.** The re-ranker can override the retriever, so covering one step is not enough. Apply the same rules to the query as well as to the documents.

4. **Audit the transform, not only the model.** The substitution list is short and readable, so someone should read it. Someone should also spot-check the output for gendered words that slipped through: school names, awards, sports, honorifics.

The guarantee is simple, and it does not require trusting the model. Once the only gender-differing words are gone, the two documents are the same text, and any ranker must score the same text the same way.

6 Limits

Our transform removes gender only from the channels it covers. Real resumes signal gender in other ways: single-sex schools, gendered awards, sports, writing style. Removing names and pronouns is therefore necessary but not sufficient in the field. Our documents are short synthetic templates rather than real applications.

The pipeline and de-identification results come from one model pair, while the 14-model audit covers only the single-document test. Our names are among the most common American given names and are mostly associated with White Americans, so we cannot separate gender from race. We treat gender as binary, which real applicants are not. The commercial models are black boxes, so we report the version strings and query dates in the appendix.

One structural point deserves care in any extension to real documents. The two stages need not read the same text. Our retriever accepts 512 tokens; our re-ranker accepts 8,192. On a long resume the two stages would therefore score different amounts of the same candidate, and some of the divergence we attribute to the re-ranker would instead be an artifact of what each stage could see. Our documents are short enough that neither model truncates anything, so the comparison here is clean. It would not be on real resumes.

7 Conclusion

Ranking models score identical men and women differently, and the difference tracks job stereotypes. Which sex a given model favors by default cannot be known without testing it. In a pool of interchangeable candidates, one common pipeline gave men 79% of shortlist places, and the re-ranker overrode the retriever. Removing names and pronouns before scoring removes the difference, because it removes the only thing that differs. The fix is cheap, checkable, and available today, and deployers should be expected to use it.

Data and code availability. The dataset generator, the model scores, the pipeline simulation, and the de-identification code are at https://github.com/vasylrakivnenko/bias_in_rerankers. Every number in this paper is produced by a script in that repository.

A How to read the numbers

Counterfactual pair. One male-named document and its female-named counterpart, scored against the same query. The two texts are identical apart from the name and the gendered pronouns.

The score gap. The man's score minus the woman's score for one counterfactual pair. Models report scores on different, incomparable scales, so we never average the gap across models and report it only within a model.

Favoring the man. The share of counterfactual pairs in which the man's score is strictly higher. Equal scores are counted as ties and reported in their own column, never credited to either side.

Shortlist. The top 3 of a pool of 20 equally qualified candidates for one job.

Occupation labels. Male-typed, female-typed, or evenly split, assigned from federal labor-force statistics on the share of women employed in each occupation.

Confidence intervals. The counterfactual pairs are not independent, because they are built from the same occupations, names, and templates. All intervals resample whole occupations rather than individual comparisons.

B Models audited

Table A1 lists every re-ranker in the audit with its vendor, how it was accessed, the date it was queried, its score range, and its tie rate. Requests to one deprecated Google model returned scores byte-identical to those of its successor, so the two are reported once, as one model. Table A2 gives the three-way split for each model and job category, and Table A3 counts how many individual occupations lean each way.

Table A1: The re-rankers audited: vendor, access route, number of counterfactual pairs, score range, share of comparisons that tied, and date scored.

Model	Vendor / access	n	Score range	Ties %	Scored
rerank-2.5	Voyage AI (API)	8,320	0.459–0.949	13.4	2026-02-24
semantic-ranker-default-002	Google Vertex AI Ranking (API)	8,320	0.258–0.684	7.7	2026-02-24
ibm-granite-granite-embedding-reranker-english-r2	IBM (via Azure AI Foundry) (API)	8,320	0.732–0.902	0.0	2026-02-24
BAAI/bge-reranker-v2-m3	BAAI (open weights) (local)	8,320	0.006–0.999	0.0	2026-02-24
rerank-2	Voyage AI (API)	8,320	0.383–0.910	18.0	2026-02-24
cross-encoder/ms-marco-MiniLM-L-12-v2	Microsoft / ms-marco (open weights) (local)	8,320	-6.599–9.535	0.0	2026-02-24
rerank-2.5-lite	Voyage AI (API)	8,320	0.459–0.926	18.5	2026-02-24
rerank-1	Voyage AI (API)	8,320	0.000–0.992	9.2	2026-02-24
Cohere-rerank-v4.0-pro	Cohere (API)	8,320	0.255–0.990	2.7	2026-02-24
semantic-ranker-fast-004	Google Vertex AI Ranking (API)	8,320	0.079–0.705	0.3	2026-02-24
Cohere-rerank-v4.0-fast	Cohere (API)	8,320	0.249–0.909	5.2	2026-02-24
semantic-ranker-default-004	Google Vertex AI Ranking (API)	8,320	0.160–0.711	0.3	2026-02-24
jinaai/jina-reranker-v2-base-multilingual	Jina AI (open weights) (local)	8,320	0.194–0.844	6.8	2026-02-24
rerank-2-lite	Voyage AI (API)	8,320	0.443–0.762	20.6	2026-02-24

Table A2: Share of counterfactual pairs favoring the man's document (M), tied, and favoring the woman's document (F), by model and job category. The last column is tie-aware: ties count as half a win each. An even-handed model would show equal M and F shares in every row.

Model (vendor)	Category	% M	% tie	% F	Tie-aware % M [95% CI]
rerank-2.5 (Voyage AI)	male	93.1	4.8	2.2	95.4 [94.3, 96.5]
	female	26.9	21.7	51.4	37.7 [27.7, 47.4]
	neutral	78.5	11.8	9.7	84.4 [78.1, 88.8]
semantic-ranker-default-002 (Google Vertex AI Ranking)	male	88.8	5.5	5.8	91.5 [87.9, 94.6]
	female	62.4	8.5	29.1	66.7 [54.2, 77.6]
	neutral	73.8	9.5	16.7	78.5 [65.1, 88.6]
ibm-granite-granite-embedding-reranker-english-r2 (IBM (via Azure AI Foundry))	male	91.3	0.0	8.7	91.3 [89.9, 92.9]
	female	76.0	0.0	24.0	76.0 [69.7, 81.5]
	neutral	87.8	0.0	12.2	87.8 [84.7, 90.7]
BAAI/bge-reranker-v2-m3 (BAAI (open weights))	male	87.0	0.0	13.0	87.0 [84.5, 89.4]
	female	71.9	0.0	28.1	71.9 [61.5, 79.8]
	neutral	75.4	0.0	24.6	75.4 [71.4, 79.1]
rerank-2 (Voyage AI)	male	80.1	11.1	8.8	85.6 [80.9, 89.8]
	female	18.2	20.4	61.4	28.4 [19.0, 37.7]
	neutral	50.1	23.4	26.5	61.8 [56.3, 67.7]
cross-encoder/ms-marco-MiniLM-L-12-v2 (Microsoft / ms-marco (open weights))	male	84.8	0.0	15.2	84.8 [79.0, 88.6]
	female	58.8	0.0	41.2	58.8 [46.1, 71.0]
	neutral	86.4	0.0	13.6	86.4 [82.3, 89.7]
rerank-2.5-lite (Voyage AI)	male	72.7	14.0	13.3	79.7 [76.0, 83.2]
	female	34.2	20.6	45.2	44.5 [38.1, 51.1]
	neutral	49.5	21.2	29.3	60.1 [54.5, 65.6]
rerank-1 (Voyage AI)	male	75.6	6.5	17.9	78.9 [72.8, 83.6]
	female	41.9	10.5	47.6	47.2 [39.2, 55.5]
	neutral	68.0	11.0	21.0	73.5 [69.4, 77.6]
Cohere-rerank-v4.0-pro (Cohere)	male	70.6	2.5	26.8	71.9 [68.4, 75.1]
	female	29.6	2.5	67.9	30.8 [23.9, 37.7]
	neutral	49.8	3.3	46.9	51.4 [46.6, 56.1]
semantic-ranker-fast-004 (Google Vertex AI Ranking)	male	67.4	0.2	32.4	67.5 [61.4, 72.1]
	female	28.7	0.2	71.0	28.8 [23.1, 35.9]
	neutral	48.2	0.4	51.4	48.4 [43.7, 53.5]
Cohere-rerank-v4.0-fast (Cohere)	male	64.7	5.5	29.8	67.5 [63.2, 71.6]
	female	33.4	5.4	61.3	36.0 [29.1, 43.6]
	neutral	62.0	4.3	33.7	64.2 [57.2, 70.7]
semantic-ranker-default-004 (Google Vertex AI Ranking)	male	59.1	0.3	40.7	59.2 [52.9, 65.8]
	female	15.5	0.2	84.3	15.6 [10.2, 22.2]
	neutral	32.6	0.3	67.1	32.8 [27.2, 38.4]
jinaai/jina-reranker-v2-base-multilingual (Jina AI (open weights))	male	45.9	6.8	47.3	49.3 [42.9, 55.4]
	female	31.8	6.5	61.7	35.1 [28.4, 40.8]
	neutral	39.1	7.1	53.8	42.7 [34.6, 50.8]
rerank-2-lite (Voyage AI)	male	34.5	22.7	42.8	45.8 [41.4, 50.1]
	female	17.3	18.7	64.0	26.7 [21.7, 32.0]
	neutral	26.9	20.7	52.5	37.2 [32.5, 42.8]

Table A3: Occupation-level consistency: how many individual occupations in each category lean in the stereotype-consistent direction, out of the number of occupations in that category.

Model	male-stereo. leaning male	female-stereo. leaning female	neutral leaning male
rerank-2.5	18/18	16/21	13/13
semantic-ranker-default-002	18/18	7/21	12/13
ibm-granite-granite-embedding-reranker-english-r2	18/18	2/21	13/13
BAAI/bge-reranker-v2-m3	18/18	3/21	13/13
rerank-2	18/18	16/21	10/13
cross-encoder/ms-marco-MiniLM-L-12-v2	17/18	6/21	13/13
rerank-2.5-lite	18/18	13/21	11/13
rerank-1	17/18	10/21	13/13
Cohere-rerank-v4.0-pro	18/18	17/21	7/13
semantic-ranker-fast-004	17/18	18/21	4/13
Cohere-rerank-v4.0-fast	17/18	14/21	12/13
semantic-ranker-default-004	13/18	20/21	0/13
jinaai/jina-reranker-v2-base-multilingual	7/18	18/21	4/13
rerank-2-lite	6/18	20/21	2/13

C Robustness

The lean varies with the wording of the document and of the search, and with which pair of names is used. Tables A4–A6 break the tie-aware share favoring the man's document down three ways. The spread across name pairs is the largest of the three, which is why deleting names alone changes the size of the disparity so much (see the third finding).

Table A4: Tie-aware share favoring the man's document, by document template.

Model	linkedin_style	news_article	professional_bio	resume_summary
rerank-2.5	77.5	65.6	64.0	70.5
semantic-ranker-default-002	76.3	73.8	71.6	91.2
ibm-granite-granite-embedding-reranker-english-r2	78.4	80.0	94.4	84.2
BAAI/bge-reranker-v2-m3	84.2	87.2	56.0	84.7
rerank-2	65.1	51.3	47.4	62.5
cross-encoder/ms-marco-MiniLM-L-12-v2	66.5	83.9	65.5	82.7
rerank-2.5-lite	32.9	70.2	59.5	79.7
rerank-1	66.1	66.6	58.8	67.3
Cohere-rerank-v4.0-pro	49.5	49.9	60.1	41.2
semantic-ranker-fast-004	40.3	46.1	58.6	43.5
Cohere-rerank-v4.0-fast	63.0	51.9	54.2	46.7
semantic-ranker-default-004	21.6	58.5	32.9	27.0
jinaai/jina-reranker-v2-base-multilingual	28.8	38.4	61.1	39.4
rerank-2-lite	41.1	51.2	27.0	24.3

Table A5: Tie-aware share favoring the man's document, by phrasing of the search.

Model	bare	experienced	expertise	qualified
rerank-2.5	68.6	76.7	68.4	63.8
semantic-ranker-default-002	75.1	78.1	78.8	80.9
ibm-granite-granite-embedding-reranker-english-r2	82.7	88.6	82.7	83.0
BAAI/bge-reranker-v2-m3	62.7	81.3	77.0	91.2
rerank-2	46.1	61.0	67.1	52.0
cross-encoder/ms-marco-MiniLM-L-12-v2	84.7	68.6	86.6	58.8
rerank-2.5-lite	54.7	60.8	56.9	69.9
rerank-1	56.8	57.8	75.2	69.0
Cohere-rerank-v4.0-pro	48.1	52.8	46.7	53.1
semantic-ranker-fast-004	39.9	72.7	41.8	34.0
Cohere-rerank-v4.0-fast	35.7	84.6	58.3	37.3
semantic-ranker-default-004	29.6	49.9	14.7	45.9
jinaai/jina-reranker-v2-base-multilingual	50.5	44.5	32.8	39.8
rerank-2-lite	23.2	45.3	44.5	30.6

Table A6: Tie-aware share favoring the man's document, by name pair.

Model	Christopher/Margaret	Daniel/Susan	David/Linda	James/Sarah	John/Maria	Michael/Jessica	Richard/Patricia	Robert/Jennifer	Thomas/Barbara	William/Elizabeth
rerank-2.5	83.8	66.6	82.2	50.5	72.5	63.8	63.6	73.6	73.6	63.5
semantic-ranker-default-002	87.1	82.5	74.0	63.2	81.0	76.4	88.5	85.0	76.9	67.4
ibm-granite-granite-embedding-reranker-english-r2	74.9	69.1	94.8	62.4	97.7	94.1	77.0	91.5	85.0	95.9
BAAI/bge-reranker-v2-m3	59.5	91.8	90.5	79.4	63.3	88.8	78.6	88.1	60.3	79.8
rerank-2	51.8	66.5	67.4	40.2	65.8	49.9	56.4	60.3	50.4	56.7
cross-encoder/ms-marco-MiniLM-L-12-v2	78.5	61.5	90.6	70.2	89.2	82.9	71.9	80.9	61.5	59.4
rerank-2.5-lite	63.9	67.8	69.7	43.6	49.3	54.3	63.8	59.8	75.1	58.5
rerank-1	59.3	52.6	85.2	40.6	88.5	66.4	79.6	77.3	38.0	59.7
Cohere-rerank-v4.0-pro	42.0	37.1	76.0	4.4	70.1	58.2	58.7	45.9	66.5	43.0
semantic-ranker-fast-004	57.6	50.8	55.8	37.7	29.3	50.6	49.0	30.0	52.0	58.1
Cohere-rerank-v4.0-fast	58.5	49.6	70.0	32.0	62.7	56.3	72.2	35.9	60.6	41.8
semantic-ranker-default-004	31.4	34.4	29.9	43.9	28.7	35.6	43.0	42.1	31.4	29.7
jinaai/jina-reranker-v2-base-multilingual	76.5	77.6	30.0	58.0	18.9	48.3	37.1	28.7	30.4	13.6
rerank-2-lite	33.4	19.8	59.6	22.5	70.4	31.3	39.7	18.4	45.9	18.4

D The same pipeline with every re-ranker

The pipeline result in the body uses one open-source pair. Table A7 repeats the simulation with each of the audited re-rankers behind the same retriever. The shortlist skew follows the re-ranker, in both size and direction, which is the point of the second finding.

Table A7: Share of shortlist places going to men's names, by re-ranker, with the same retriever and the same pool of equally qualified candidates.

Re-ranker	Vendor	Pipeline % male in top-3				rr-alone	rank gap
		all	male	female	neutral	all	all
ibm-granite-granite-embedding-reranker-english-r2	IBM (via Azure AI Foundry)	84.9	95.9	71.7	90.7	89.3	3.69
semantic-ranker-default-002	Google Vertex AI Ranking	79.9	94.4	63.1	86.7	77.5	3.48
BAAI/bge-reranker-v2-m3	BAAI (open weights)	79.3	95.0	67.3	76.9	78.6	3.41
cross-encoder/ms-marco-MiniLM-L-12-v2	Microsoft / ms-marco (open weights)	78.4	91.9	58.6	91.7	78.6	3.32
rerank-1	Voyage AI	70.5	91.1	45.1	82.8	68.3	2.94
rerank-2.5	Voyage AI	70.2	98.1	36.6	85.9	71.7	3.00
rerank-2.5-lite	Voyage AI	62.7	88.6	39.1	64.8	56.9	2.62
rerank-2	Voyage AI	59.8	91.2	29.7	65.1	57.2	2.47
Cohere-rerank-v4.0-fast	Cohere	56.6	77.6	33.1	65.5	47.4	2.42
semantic-ranker-fast-004	Google Vertex AI Ranking	56.2	84.0	31.2	58.1	47.6	2.19
Cohere-rerank-v4.0-pro	Cohere	51.7	74.8	31.9	51.6	42.6	2.14
jinaai/jina-reranker-v2-base-multilingual	Jina AI (open weights)	50.1	67.8	34.2	51.2	31.2	2.03
rerank-2-lite	Voyage AI	45.1	66.5	26.2	46.1	38.4	1.69
semantic-ranker-default-004	Google Vertex AI Ranking	44.2	77.5	17.3	41.6	31.7	1.71

E De-identification conditions

Table A8 reports every de-identification rule tested on the same re-ranker, including each half of the transform on its own and the grammatical rewrites. The full transform makes the two documents byte-identical, so its result follows by construction; it is reported as a check on the symmetry argument, not as a measurement.

Table A8: De-identification conditions: the share of counterfactual pairs favoring the man's document, the share tied, and the average size of the score gap on this model's own scale.

Condition	% male-favoured	% ties	mean |score gap|	twins identical %	top-1 %
Original (no transform)	78.0	0.0	0.0358	0.0	99.5
Names only	93.3	0.0	0.0334	0.0	98.6
Pronouns only	46.9	0.0	0.0239	0.0	98.1
Full (names + pronouns → one token)†	50.0	100.0	0.0000	100.0	99.5
Grammatical, naive ('her' → 'their')	37.5	75.0	0.0009	75.0	–
Grammatical, naive ('her' → 'them')	87.0	0.0	0.0075	0.0	–
Grammatical, POS-aware†	50.0	100.0	0.0000	100.0	99.5

References

[1] J. An, D. Huang, C. Lin, and M. Tai. Measuring gender and racial biases in large language models: Intersectional evidence from automated resume evaluation. *PNAS Nexus*, 4(3):pgaf089, 2025. doi:10.1093/pnasnexus/pgaf089.

[2] Bureau of Labor Statistics, U.S. Department of Labor. Employed people by detailed occupation, sex, race, and Hispanic or Latino ethnicity. Current Population Survey, Household Data Annual Averages, Table 11, 2025 annual averages. `https://www.bls.gov/cps/cpsaat11.htm` (accessed August 2026).

[3] A. Caliskan, J. J. Bryson, and A. Narayanan. Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334):183–186, 2017. doi:10.1126/science.aal4230.

[4] Civil Rights Act of 1964, Title VII, 42 U.S.C. §§ 2000e *et seq.* Disparate impact codified at 42 U.S.C. § 2000e-2(k)(1)(A), added by the Civil Rights Act of 1991, Pub. L. No. 102-166, § 105(a), 105 Stat. 1074.

[5] Google Cloud. Cloud Talent Solution: Job Search overview. `https://docs.cloud.google.com/talent-solution/job-search/docs` (accessed August 2026).

[6] *Griggs v. Duke Power Co.*, 401 U.S. 424 (1971).

[7] Y. Hou, X. Pan, W. X. Zhao, S. Bian, Y. Song, T. Zhang, and J.-R. Wen. Leveraging Search History for Improving Person-Job Fit. In *Database Systems for Advanced Applications (DASFAA 2022)*, Lecture Notes in Computer Science 13245, pages 38–54. Springer, 2022. doi:10.1007/978-3-031-00123-9_3. arXiv:2203.14232.

[8] Indeed Engineering. Distilling Long-Tail User Behavior into Scalable Embeddings for Job Search. `https://engineering.indeedblog.com/blog/2026/06/distilling-long-tail-user-behavior-into-scalable-embeddings-for-job-search/` (accessed June 2026).

[9] LinkedIn Engineering. The AI Behind LinkedIn Recruiter search and recommendation systems. Q. Guo, S. C. Geyik, C. Ozcaglar, K. Thakkar, N. Anjum, and K. Kenthapadi, 22 April 2019. `https://www.linkedin.com/blog/engineering/recommendations/ai-behind-linkedin-recruiter-search-and-recommendation-systems` (accessed August 2026).

[10] R. Nogueira and K. Cho. Passage Re-ranking with BERT. arXiv preprint arXiv:1901.04085, 2019.

[11] V. Rakivnenko, N. Maslej, J. Cervi, and V. Zhukov. Bias in Text Embedding Models. arXiv preprint arXiv:2406.12138, 2024.

[12] Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 (Artificial Intelligence Act), OJ L, 2024/1689, 12.7.2024. Annex III, point 4(a); Recital 57; Arts. 16(a), 26 and 27.

[13] Regulation (EU) 2026/1744 of the European Parliament and of the Council of 8 July 2026 (Digital Omnibus on AI), OJ L, 2026/1744, 24.7.2026. Amending Art. 113 of Regulation (EU) 2024/1689: Chapter III, Sections 1–3 apply from 2 December 2027 to high-risk AI systems under Art. 6(2) and Annex III.

[14] N. Rekabsaz and M. Schedl. Do Neural Ranking Models Intensify Gender Bias? In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2065–2068, 2020. doi:10.1145/3397271.3401280. arXiv:2005.00372.

[15] N. Rekabsaz, S. Kopeinik, and M. Schedl. Societal Biases in Retrieved Contents: Measurement Framework and Adversarial Mitigation of BERT Rankers. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 306–316, 2021. doi:10.1145/3404835.3462949. arXiv:2104.13640.

[16] Social Security Administration. National data on the relative frequency of given names in the population of U.S. births. Tabulated from Social Security records as of 1 March 2026. `https://www.ssa.gov/oact/babynames/limits.html` (accessed August 2026).

[17] Uniform Guidelines on Employee Selection Procedures (1978), 29 C.F.R. pt. 1607. Four-fifths rule at § 1607.4(D).

[18] *Watson v. Fort Worth Bank & Trust*, 487 U.S. 977, 998 (1988) (plurality opinion).

[19] K. Wilson and A. Caliskan. Gender, Race, and Intersectional Bias in Resume Screening via Language Model Retrieval. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 7(1):1578–1590, 2024. doi:10.1609/aies.v7i1.31748.

[20] X. Yu, R. Xu, C. Xue, J. Zhang, X. Ma, and Z. Yu. ConFit v2: Improving Resume-Job Matching using Hypothetical Resume Embedding and Runner-Up Hard-Negative Mining. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 12775–12790, 2025. doi:10.18653/v1/2025.findings-acl.661. arXiv:2502.12361.